P. Krishna Chaitanya

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PROFILE

Undergraduate physics student (Honours with Research) with hands-on experience in solid-state synthesis and X-ray characterisation of piezoelectric ceramics, solid rocket motor design and testing, and aircraft stability modelling. Pursuing a Flight Mechanics minor alongside the physics degree.

EDUCATION

Sri Sathya Sai Institute of Higher Learning (SSSIHL) — Prasanthi Nilayam, India

B.Sc. (Hons.) Physics with Research | 2025 – Expected 2029

- CGPA: 7.2 / 10 (cumulative, Semesters 1–2)
- Minor: Flight Mechanics — aerodynamics; longitudinal, directional and lateral stability; stability derivatives
- Coursework: Classical Mechanics, Electromagnetism, Quantum Mechanics, Mathematical Physics, Electronics, Computational Techniques in Physics

RESEARCH EXPERIENCE

Research Intern — School of Engineering Sciences & Technology (SEST), University of Hyderabad

Under Prof. Dibakar Das | 18 May – 18 June 2026 | Hyderabad, India

- Synthesised lead-free $K_{0.48}Na_{0.52}NbO_3$ (KNN) piezoceramics by the solid-state reaction route, carrying out the synthesis, calcination, pellet pressing and sintering myself.
- Compared two calcination schedules (825 °C / 4 h and 900 °C / 2 h); XRD confirmed a single-phase orthorhombic perovskite with a minor $K_2Nb_4O_{11}$ secondary phase.
- Raised sintered relative density from about 64% at 1000 °C to about 91% at 1100 °C / 4 h.

PROJECTS

KNSU Solid Rocket Motor — Design, Fabrication and Test (ongoing)

- Designing and building a sugar-propellant (KNSU) motor with an aluminium casing, flown in a PVC airframe with through-the-wall fins.
- First launch attempt did not lift off; diagnosed insufficient thrust as the cause.
- Redesigned the nozzle for a scaled 30 mm variant (70:30 plaster-of-Paris/cement composite with a metal throat insert), with STL models and a written design basis.
- Next: an instrumented static fire to measure the thrust curve, and a CP/CG stability check before the next flight.

Aerodynamic Modelling and Stability Analysis

- Built aircraft geometry models in OpenVSP and ran VSPAERO analyses to extract stability derivatives, focusing on the vertical tail's contribution to directional stability.
- Completed all 30 lectures of the NPTEL course *Aircraft Stability and Control* (Prof. A. K. Ghosh, IIT Kanpur); certification examination on 25 October 2026.

Computational Physics in Python

- Implemented a finite-difference solver for the lid-driven cavity problem in computational fluid dynamics, and simulations of projectile motion and fluid behaviour, using NumPy, SciPy and Matplotlib.
- Built an automated laboratory-record pipeline (ReportLab and Matplotlib) generating print-ready reports and plots directly from measured data.
- Wrote a command-line habit-tracking application in Python.

AI-Directed Software Builds

- Directed AI coding agents through multi-version builds: a Unity 6 game prologue (combat, enemies, boss fight, cutscenes), an always-on Windows voice assistant (speech-to-text, OCR, app control) and a webcam hand-tracking control app.

ACADEMIC PRESENTATION

Poster Presentation — NUPC 2025

Photobiomodulation and glucose modulation under red-light exposure

- Reviewed the literature and organised the scientific background into a structured poster covering motivation, mechanism, key observations and future scope.

LABORATORY EXPERIENCE

- **Electronics (Analog and Digital):** circuit analysis and measurement, standard laboratory instrumentation, interpretation of experimental observations.
- **Mechanics:** measurement, data collection, graphical analysis and error analysis.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, ReportLab) — numerical simulation, data analysis and automated report generation.
- **Scientific and engineering tools:** OpenVSP / VSPAERO, Scilab, Excel / Google Sheets, AI coding agents (Claude Code).
- **Experimental:** solid-state synthesis, pellet pressing and sintering, density measurement, X-ray diffraction, analog and digital electronics, error analysis.
- **Engineering:** aerodynamics and flight stability, propulsion fundamentals, fabrication and prototyping.
- **Communication:** literature review, technical writing, poster presentation.
- **Languages:** English — fluent (medium of instruction); German — A2 in progress, working towards B1.